

Comparative analysis of piezo electric and accelerometer sensor for the design of rate adaptive pacemaker

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ABSTRACT

This paper presents a comparative study of activity sensors: piezoelectric and accelerometer sensor, for the design of rate adaptive pacemaker. Activity sensors are incorporated with the demand mode pacemaker for achieving rate adaptiveness of the pacing. There is a conflict in selecting the activity sensors: accelerometer and piezoelectric. These sensors could monitor the physical activity of the patients and adapt the pacing rate to the needs of the activity of the patient. Implanting these physiological sensors into pacemakers made it possible for the physicians to individually optimize pacing parameters for each patient. Piezoelectric sensors may not response accurately for some kind of activities like cycling, walking down stairs etc. Accelerometer sensors have also the limitations for measuring prolonged duration activities. Clinically, accelerometer based activity sensors have dominated the piezoelectric based activity sensor for the design of rate adaptive pacemaker. Therefore, in this paper we have performed the frequency and time domain analysis for analyzing the technically suitable activity sensor for the design of rate adaptive pacemaker and it is found that accelerometer performs 90% faster than piezoelectric sensor.

1. Introduction

Cardiovascular Diseases (CVDs) are still leading cause of death worldwide, approximately 31% of overall death cases [1]. The increasing rate of cardiovascular diseases causing untimely deaths, has made increase in need of implantable cardiac devices such as: pacemaker and defibrillators. With the advent of the sixth decade of implantable cardiac pacemakers, the demand for automaticity in pacers has become increasingly perceptible. Basically pacemakers are of three types: Fixed Rate Pacemaker, Demand Pacemaker and Rate Adaptive Pacemaker [2]. Fixed rate pacemakers are the pacemakers which generates external stimulus at fixed duration to provide the pulse rate at 72 beats per minute (bpm). Demand pacemakers are the pacemakers which delivers stimulus whenever needed by the bradycardiac patient and the Rate adaptive pacemakers are the pacemaker which can vary the stimulus delivery rate according to the physiological activity of the bradycardiac patient. Rate adaptive pacing in bradycardia patients is due to chronic atrial fibrillation and a slow ventricular response.

Recent developments in the design of rate adaptive pacemaker comprises the use of various sensors such as: activity sensor, minute

ventilation sensor, impedance sensor, oxygen saturation sensor, pressure sensor etc. [3–8]. Rate adaptive pacemakers are designed by comprising activity sensor in demand mode pacemakers. Ideally, there should be no delay between the need of the pacing to the heart and the stimulus delivery by the pacemaker. For achieving this goal, activity sensors are selected to provide accurate result. Basic block diagram of rate adaptive pacemaker is shown in Fig. 1.

Fig. 1 presents the basic block diagram of rate adaptive pacemaker, comprises of sensing unit, pacing rate controller, excitation signal generator and finally the patient's activity sensor. Excluding the activity sensor, the rest is the basic block diagram of demand pacemaker. Sensing unit senses the internal pacing of the bradycardiac patient, Excitation signal generator delivers the stimulus to the muscles of the heart and Pacing rate controller controls the duration of the pulse to be delivered according to the activity sensor's response. Activity sensor either be a piezoelectric sensor or accelerometer sensor.

Piezoelectric (PEZ) sensor operates on threshold based operation but sometimes it fails to judge the exact physiological activity such as: cycling and walking downstairs. To overcome these limitation, accelerometer sensors are used. Accelerometer (ACC) sensor has also the

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limitation in delivering constant stimulus for prolong activity as acceleration remains constant in this case. Several literatures are available for providing theoretical analysis regarding selection of activity sensor [3–16]. Therefore, through this paper, we will discuss the frequency domain and time domain analysis using transfer function approach for Piezoelectric and Accelerometer sensor. The rest of the paper is as follows: section 2 presents Material and methods for selecting PEZ and ACC sensor. Section 2 describes theory of simulation based approach of the proposed paper. Section 3 discusses the result. Discussion and conclusion are presented in section 4 and section 5 respectively.

2. Material and methods

Rate Adaptive (RA) pacing can be categorized in two classes: Open Loop RA Pacing and Closed Loop RA Pacing. In the open-loop design, the activity sensor's output is translated to change in pacing rate according to the pacing algorithm. As it is an open loop design, there will not be any feedback from the pacemaker algorithm to the sensor's parameter. In the closed-loop design, the rate change resulting from sensor activation results in a change in the physiologic parameter in the opposite direction, i.e., a negative feedback loop. In the open loop RA pacing, rate of change of stimulus delivery is decided by clinician and in closed loop RA pacing the rate of change of stimulus delivery is decided according to the mental and physical stress of the bradycardiac patient. Block diagram of open loop and closed loop pacing is shown in Fig. 2.

Industrial level design of pacemaker follows the closed loop rate adaptive pacing system. The activity sensor is selected for the Rate Adaptive Pacemaker (RAPM) design which are easily implantable, requires less space and most important, response as quick as possible (ideally without any delay). Nowadays, leadless RA pacemakers are also available [17–19] and there is still a conflict in selecting the appropriate activity sensor: piezoelectric or accelerometer sensor as both the sensors have certain limitations. PEZ sensors are generally mounted at the surface of the rate adaptive leadless pacemaker can, whereas ACC sensors are placed with the pulse generator circuitry. The main difference in the operation of PEZ and ACC is, PEZ sensor responses at threshold based approach, shown in Fig. 3, whereas ACC sensor responses by integrating the voltage signal which is acquired from the sensor according to the physiological activity of the patient.

In Fig. 3, the threshold value for PEZ sensor is set to 4, above which increase in pacing rate will be required. In some of the PEZ sensors, this threshold value is sub classified into three categories: low, medium and high, to adapt more accurate change in pacing rate. The next difference between the two sensors is, PEZ performs according to the vibrations accessed by the surface of the RA pacemaker however ACC performs according to the accelerations received by the sensor. Apart from the theoretical analysis, we have performed some Simulation analysis for these two sensors, to attain more appropriate results for selecting the

more suitable activity sensor for RAPM design. The detail of selecting transfer function and further analysis are discussed in the next section.

3. Theory/calculation

For technical analysis of response of activity sensor, MATLAB/Simulink has been used here. First of all, transfer function has been selected for PEZ and ACC according to spring damping analysis. It has been mentioned that the schematic of a leadless pacemaker can be represented in the form of a spring damping system [9]. Several literature are available regarding transfer function of PEZ and ACC sensor using spring damping system [20–29]. The transfer function is the easiest way to analyze any system, with the help of which we can easily perform open loop and closed loop analysis. The transfer function of PEZ [30] and ACC [20] using spring damping system are presented in equ.1 and equ.2.

$$PEZ = \frac{1}{0.15s^2 + 2.677s + 56} \quad (1)$$

$$ACC = \frac{1}{3.6 \times 10^{-10}s^2 + 6.25 \times 10^{-6}s + 571.5} \quad (2)$$

Behrens et al. [30] have presented passive vibration control for piezoelectric sensor and they have selected the parameters: mass (m), damping coefficient (b) and spring coefficient (k) equal to 0.15 kg, 2.677 N/m and 56 N/m respectively. Jacob et al. [20] have presented second order system analysis for accelerometer and they have selected the above mentioned parameters m, b and k equal to 3.6×10^{-10} kg, 6.25×10^{-6} Ns/m and 571.5 N/m respectively. Using these transfer functions, time and frequency domain analysis have been performed and discussed in the 'Simulation Results' and 'Discussion' sections of the manuscript.

Simulation Results: For time domain analysis, step response of the transfer function of PEZ and ACC has been obtained. The step response for the above mentioned two sensors have been shown in Fig. 4 and Fig. 5 respectively.

All the step response parameters such as Rise time, Settling time, Overshoot and Stability, have been discussed in the next section of the manuscript. After the time domain analysis, frequency domain analyses: Bode and Nyquist Plot, have been performed. The frequency domain analysis is performed to observe the change in magnitude and phase angle of the response by varying frequency from zero to infinity The Bode plot of any system is an analysis of its Gain and Phase response, with or without feedback. The Bode plots for PEZ and ACC are shown below in Fig. 6 and Fig. 7 respectively.

Further, complex Gain analysis with Phase response, for the sensors with feedback has been obtained using Nyquist Plot. The Nyquist plot also provides the information about closed loop stability of the system. The Nyquist plots for PEZ and ACC are shown below in Fig. 8 and Fig. 9

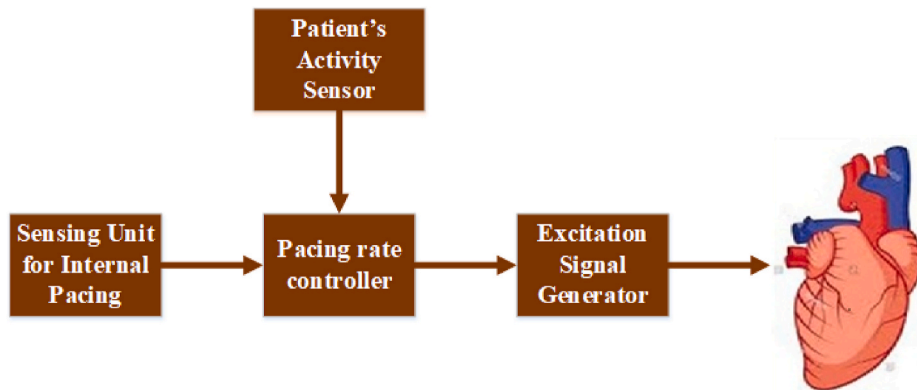


Fig. 1. Block diagram of rate adaptive pacemaker [9].

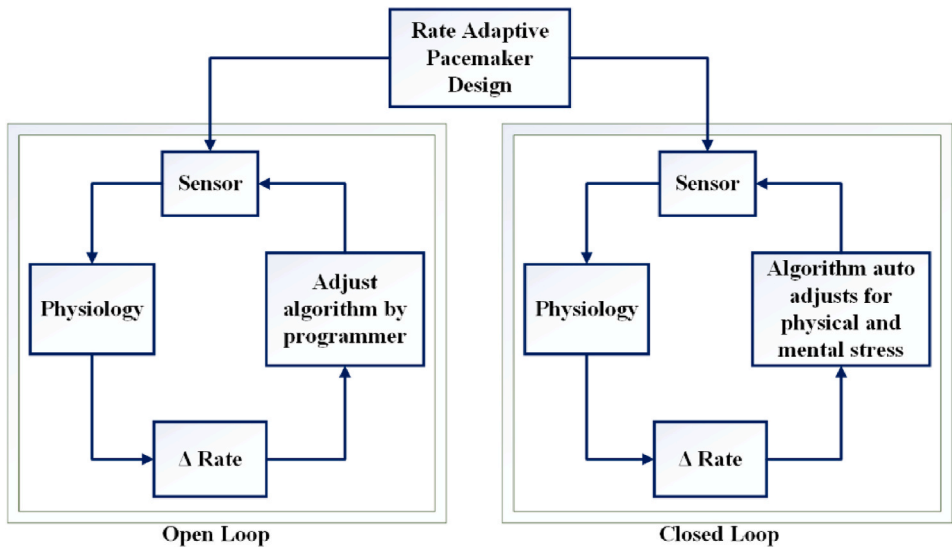


Fig. 2. Open and closed loop rate adaptive pacing system.

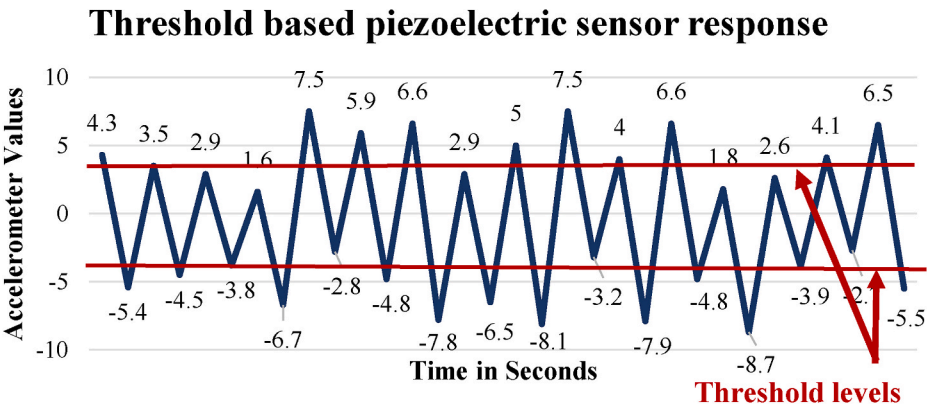


Fig. 3. Threshold based PEZ sensor response.

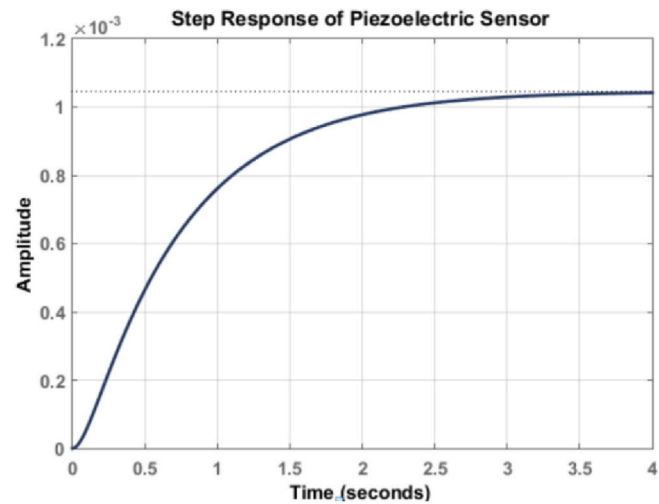


Fig. 4. Step response of PEZ Sensor.

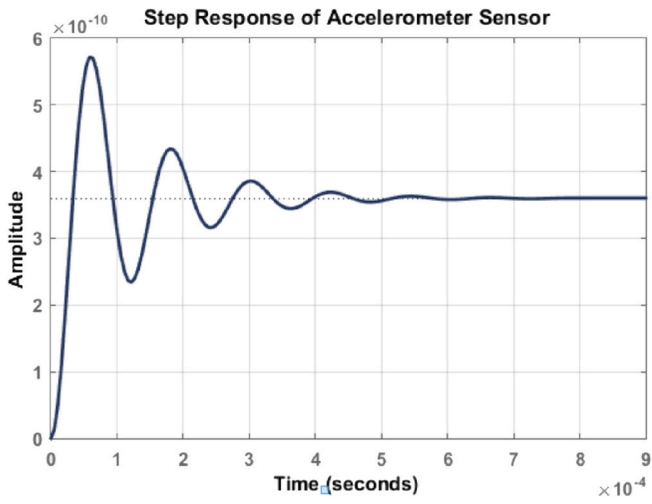


Fig. 5. Step response of ACC Sensor.

respectively. Analyses of the Step Response, Bode Plot and Nyquist Plot for PEZ and ACC sensors have been discussed in the next section of the manuscript.

4. Discussion

By analyzing the Step Response of the two sensors: PEZ and ACC, all the parameters are compared below in Table .1.

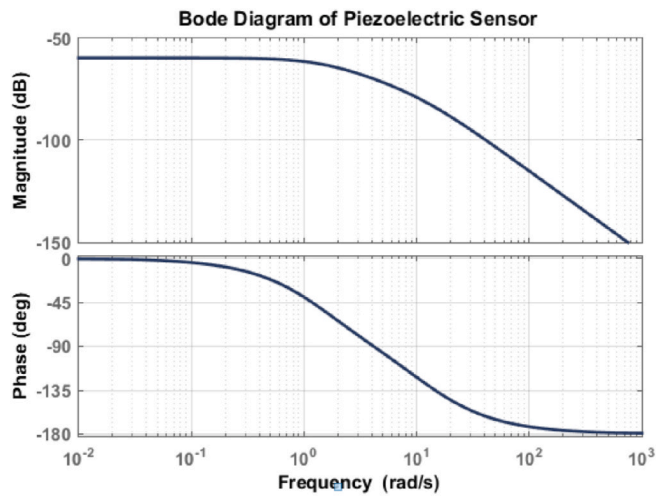


Fig. 6. Bode plot of PEZ Sensor.

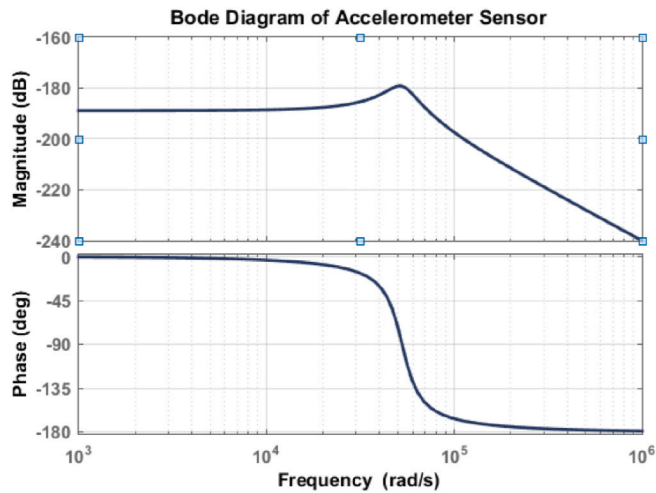


Fig. 7. Bode plot of ACC Sensor.

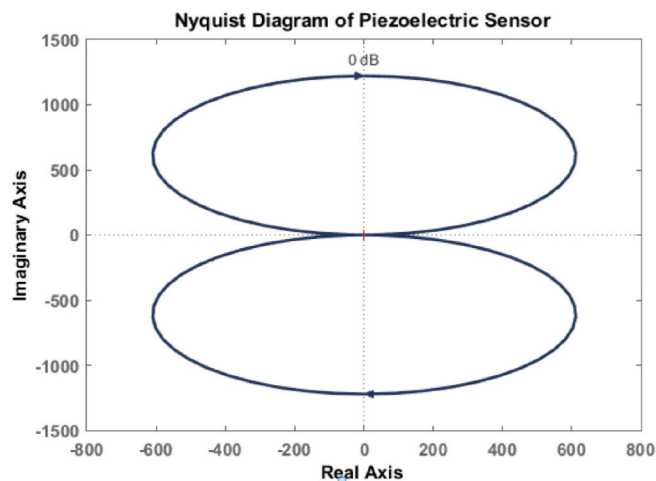


Fig. 8. Nyquist plot of PEZ Sensor.

By analyzing the step response parameters, it is obvious that all parameters: Rise Time, Settling Time, Peak Overshoot and Peak Time, of PEZ sensor are much lesser than ACC sensor, which means the ACC will

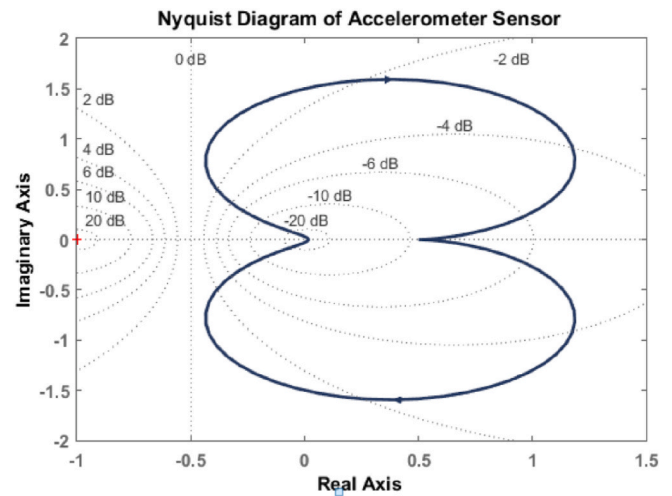


Fig. 9. Nyquist plot of ACC Sensor.

Table 1

Step response analysis of PEZ and ACC sensors.

S.No.	Parameters	PEZ	ACC
1.	Rise Time (in Seconds)	1.5587	0.0152
2.	Settling Time (in Seconds)	2.8311	0.0798
4.	Peak Overshoot	0.0010	0.0002
5.	Peak Time	4.5335	0.0393
6.	Stability	Stable	Stable

provide much faster response as compare to PEZ sensor. The analysis of Bode plots for PEZ and ACC are also analyzed and the analysis is shown in Table 2.

It can be seen from Table 2, that the Bode Plot analysis is same for the two sensor. While analyzing the Nyquist plot, it is found that the frequency domain response for both the sensor is same and stable with the feedback also. As transfer function for ACC and PEZ may vary according to the system, therefore values shown in Table 2 for frequency domain analysis cannot be conclusive. Thus, by analyzing the time and frequency domain response, it is technically proven that ACC would be a better activity sensor for pacemaker design as compare to PEZ sensor.

5. Conclusion

The paper presented a comparative study and simulation analysis of Piezoelectric and Accelerometer sensors for the design of Rate Adaptive Pacemaker. Theoretical analysis was performed and from which, the transfer function for both the sensors have been obtained. Using those transfer functions, time domain and frequency domain analyses were performed using MATLAB. It was observed that the frequency domain analyses (Bode and Nyquist Plots) are same for both the sensors, whereas from time domain analysis, it can be concluded that, ACC sensor are much faster sensors as compared to PEZ sensors for the design of rate adaptive pacemakers, which was theoretically proven in many literature.

Table 2

Bode Plot analysis of PEZ and ACC sensors.

S.No.	Parameters	PEZ	ACC
1.	Gain Margin	Infinite	Infinite
2.	Phase Margin	Infinite	Infinite
3.	Gain Crossover Frequency	Infinite	Infinite
4.	Phase Crossover Frequency	Not Applicable	Not Applicable

CRedit authorship contribution statement

Rohini Srivastava: simulated the concept, implement the concept on the software and wrote the whole manuscript, is the corresponding author as well. All authors revised the manuscript. **Nitin Sahai:** and **Ravi P. Tewari:** have visualized and analyzed the data with advice from. **Basant Kumar:** conceptualized this study and supervised the implementation, is the corresponding author as well. All authors revised the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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